

Improving Information Extraction with Learned Queries

Omar Sharif, Soroush Vosoughi, Nikhil Singh

Department of Computer Science

Dartmouth College

{omar.sharif.gr, soroush.vosoughi, nikhil.u.singh}@dartmouth.edu

Abstract

When information extraction fails, a natural instinct is to improve the model doing it: for example, by scaling it up or refining its reasoning. In this paper, we show that another part of the pipeline matters at least as much: the queries used to elicit this information. Across four clinical benchmarks and five LLMs, improving the question design alone raises performance by ≈ 18.6 F1-score points, i.e. more than using larger extraction models. To make such *question design* learnable, we introduce **List of Questions (LOQ)**, which generates document-specific question sets, and FEEDQ, a feedback-driven optimization method that iteratively refines questions against extraction outcomes. The resulting optimized questions can be used to train lightweight generators: with fine-tuning, 4B-parameter models match or outperform expert-derived baselines and substantially exceed the performance of much larger untuned models. We release a dataset of 12,825 optimized questions to support a broader shift in information extraction research toward treating question design as a first-class problem.

1 Introduction

“If you do not know how to ask the right question, you discover nothing.”

— W. Edwards Deming

We are surrounded by documents that contain answers to questions we have not yet asked. When those questions eventually arise, for example “Which medication was prescribed in each of these health records?” the task becomes one of *information extraction*: identifying and structuring the relevant pieces of information embedded in the text. Straightforward as this goal may appear, it can be surprisingly difficult. The information of interest is rarely presented explicitly or in a consistent form; rather, it is often distributed across sentences, expressed indirectly, or intertwined with

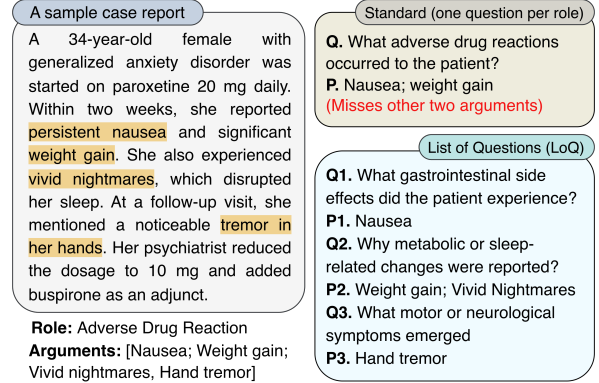


Figure 1: A case report describes a patient who develops four adverse drug reactions after starting paroxetine (highlighted). The standard approach asks a single role-level question and retrieves only two. LOQ generates a set of document-grounded questions, each targeting a different symptom category, and recovers all four. The extraction model is identical in both settings; only the questions change, showing how extracting answers benefits from formulating the right questions. Additional examples are given in Table 12.

narrative context. Thus, to get the *right* answers, we must ask the right questions.

This challenge is especially acute in clinical and biomedical text. Case reports, pharmacovigilance records, and discharge summaries convey structured facts through unstructured narratives (Zhang et al., 2025; Builtjes et al., 2025). Consider extracting just one type of information: “Adverse Drug Reaction”. Symptoms may be mentioned in one paragraph, a cause implied through temporal proximity to a medication, and a resolution described later still. Extracting such information reliably depends on both whether a model is capable and whether it is queried in a way that exposes the relevant evidence.

Large language models (LLMs) have become the dominant method for such tasks (Shimizu et al., 2025; Tanev et al., 2025). For LLMs this problem is often framed as question-based argument

extraction: given a document and question about target role, the model generates the relevant arguments as free text (Sharif et al., 2024; Zhang et al., 2024). Here, a role is the *type* of information to be extracted (e.g., *Adverse Drug Reaction*), and arguments are the specific details for that role (e.g., the drug involved, the symptoms observed, the outcome). This formulation is flexible but it quietly inherits an assumption that the questions themselves are adequate. The dominant approaches derive questions from annotation schema or role-level templates and apply them uniformly across documents regardless of how the target information is expressed (Du and Cardie, 2020; Sharif et al., 2025). As shown in Figure 1, asking “*What adverse drug reaction occurred to the patient?*” treats the role as a single retrieval target, assuming all relevant arguments can be surfaced with one generic query. This risks systematically missing arguments that require targeted attention. The bottleneck, then, is not only the LLMs’ ability to answer but the way we ask question and direct this attention.

We introduce **List of Questions (LOQ)**, a **framework for learning extraction questions tailored to each document–role pair**. LOQ uses FEEDQ, an iterative loop that refines candidate questions based on the quality of the extractions they elicit. Optimized questions then help train a lightweight question generation model for deployment. Our contributions are thus:

- We provide **empirical evidence** that question design is a major bottleneck in LLM-based information extraction. Across four clinical benchmarks (CaseReportBench (Zhang et al., 2025), PHEE (Sun et al., 2022), DiscourseEE (Sharif et al., 2024), and MACCROBAT (Ma et al., 2023)) and five LLMs, improving questions can increase F1 by **18.6%** on average.
- We introduce a **two-part framework**, LOQ, for learning to generate effective, document-grounded extraction questions:
 1. First, we develop FEEDQ, a **feedback-driven iterative refinement method** for optimizing extraction questions without annotated question supervision. We use FEEDQ to construct a **dataset of 12,825 (document, role, question-set) triples**.
 2. Second, we apply this dataset via fine-tuning and **demonstrate significant gains in zero-shot extraction**.

2 Related Work

Information extraction as question answering.

Information extraction can be straightforwardly recast as question answering (Du and Cardie, 2020; Liu et al., 2020; Li et al., 2020) by asking models a question whose answer is the desired argument. This prompts extraction models to use the question-answering and reading-comprehension abilities already present in pretrained LMs, and supports free-text rather than span-only approaches (Huang et al., 2024; Sharif et al., 2024). Once framed this way, question quality becomes instrumental to extraction success. In practice, questions are usually written once from annotation guidelines or role templates and then reused across all documents (Hsu et al., 2022; Srivastava et al., 2025). That makes them easy to apply, but also indifferent to how evidence is expressed in different documents.

Quality of questions. Recent work has started to take question formulation more seriously. Lu et al. (2023) generate better role questions from event-specific templates, Hong and Liu (2024) refine question generation with reinforcement learning, and Uddin et al. (2024a) show benefits from combining contextualized and uncontextualized questions. Despite this progress, most approaches still operate on a question-per-role paradigm (Uddin et al., 2024b), assuming each role is a single, clean retrieval target. However, a role may involve several arguments, be expressed indirectly, or scattered across sections. Current methods also usually improve question quality using proxies like template supervision (Lu et al., 2023), fluency and context-recovery rewards (Hong and Liu, 2024), or heuristic combinations of question types (Uddin et al., 2024a). Our work builds on these ideas, but instead studies question design as a separable component of extraction. **We ask: for a document-role pair, what set of questions best exposes the relevant evidence and how can we find these?**

Learning to ask the right questions. *Prompt optimization* is an emerging strategy for improving textual artifacts (Pryzant et al., 2023; Yuksekgonul et al., 2025; Lee et al., 2025; Cherep et al., 2026). Such techniques approximate gradient-based optimization via natural language *feedback*. However, these methods optimize a single text prompt, whereas our setting requires optimizing a document- and role-conditioned *set* of questions *under a multi-tier extraction objective*, with set-level

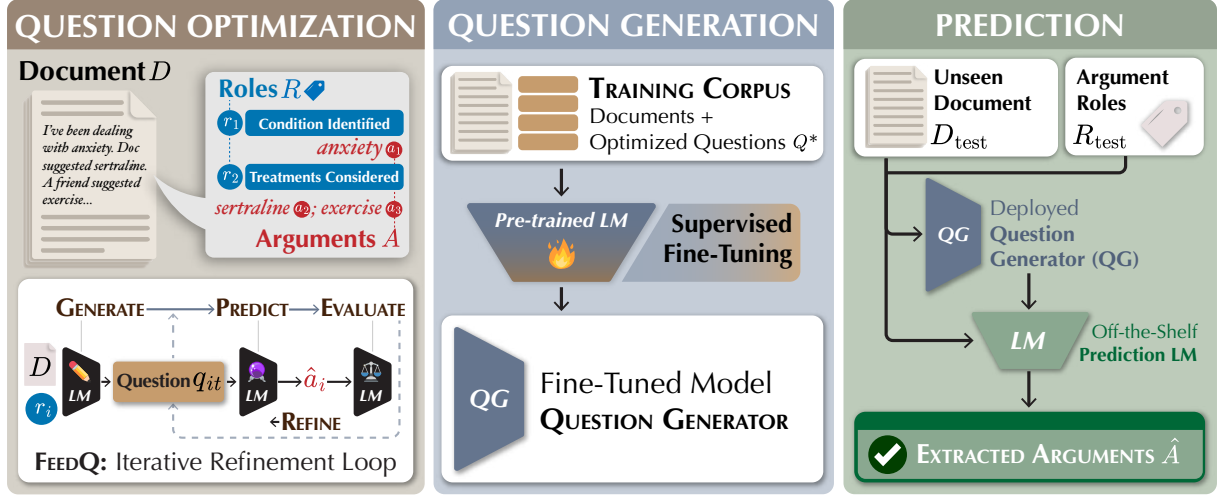


Figure 2: Overview of the List of Questions framework. (1) **Question optimization**: FEEDQ iteratively generates, evaluates, and refines role-specific questions conditioned on the document, roles, and ground-truth arguments, producing the right questions. (2) **QG fine-tuning**: These questions teach a smaller model to produce questions from documents and roles, removing the dependence on ground-truth. (3) **Prediction**: At test time, the fine-tuned QG model generates questions for unseen documents, and a prediction model uses them to extract arguments.

edits and a leakage constraint. We develop a new feedback-based optimizer which alternates question generation and information extraction, along with a multi-step evaluation and, document- and role-conditioning, a leakage check module, among several other components. We use this method to iteratively refine questions to maximize extraction quality. We then transfer this knowledge via supervised fine-tuning to a lightweight question generation LM, which produces document-specific questions *without* access to ground-truth arguments. Answering them with an off-the-shelf prediction LM should then yield, on average, better information extraction.

3 Methods

We propose the **List of Questions (LOQ)** framework, which automates the full argument extraction cycle in three phases (illustrated in Figure 2).

Phase 1: Question Optimization. Given a document, role, and ground-truth arguments, we run an iterative optimization loop (FEEDQ, Section 3.2) to derive questions that best extract those arguments, producing gold question sets for training.

Phase 2: Question Generation. We fine-tune a question generator (QG) model on these gold questions, thereby teaching it to generate effective questions conditioned on document and role.

Phase 3: Prediction. At inference, the fine-tuned QG generates questions for unseen documents, and off-the-shelf LMs use these to extract arguments.

Our design separates question generation from prediction, allowing each component to specialize.

3.1 Preliminaries

A **document** $\mathcal{D}$ is a full-length text taken from diverse clinical sources, such as case reports, pharmacovigilance records, or online health forums (e.g., reddit posts). Each $\mathcal{D}$ has an information structure defined by a set of **roles** $\mathcal{R} = \{r_1, r_2, \dots, r_n\}$ that specify the *types* of information to extract. **Arguments** $\mathcal{A} = \{a_1, a_2, \dots, a_m\}$ are the corresponding details for each of these roles. For example, given the sentence “*Patient was prescribed Metformin 500mg twice daily,*” the relevant information structure is a **MEDICATION** event with roles *drug*, *dosage*, and *frequency*, filled by the arguments *Metformin*, *500mg*, and *twice daily*, respectively. We formulate clinical information extraction as a generative argument extraction task: given a document $\mathcal{D}$ and a set of target roles $\mathcal{R}$, an LM extracts a set of arguments $\mathcal{A}$ corresponding to each role.

3.2 Question Optimization via Feedback

We design **FEEDQ**, an iterative Feedback-Driven Question Optimization approach with the goal of automatically derive the right set of questions to extract the information from a given document.

Objective. Given a document $\mathcal{D}$, a role $r \in \mathcal{R}$, and ground-truth arguments $\mathcal{A} = \{a_1, \dots, a_m\}$ for r , we seek a question set $\mathcal{Q}^*$ that maximizes

Algorithm 1 FEEDQ: Feedback-Driven Question Optimization

Require: Ground-truth arguments $\mathcal{A}$, initial questions $\mathcal{Q}^{(0)}$, document $\mathcal{D}$, role r , max iterations T , target score τ , patience m

Ensure: Optimized question set $\mathcal{Q}^*$

```

1:  $\mathcal{Q}^* \leftarrow \mathcal{Q}^{(0)}$ ;  $\tau^* \leftarrow 0.0$ ;  $p \leftarrow 0$  ▷ Initialization
2: for  $t = 0, 1, \dots, T-1$  do
3:    $\hat{\mathcal{Q}}^{(t)} \leftarrow \text{LEAKAGECHECK}(\mathcal{Q}^{(t)}, \mathcal{A})$ 
4:    $\hat{\mathcal{A}}^{(t)} \leftarrow \text{EXTRACT}(\mathcal{D}, r, \hat{\mathcal{Q}}^{(t)})$  ▷ Prediction
   // Evaluation
5:    $M^{(t)} \leftarrow \hat{\mathcal{A}}^{(t)} \cap \mathcal{A}$ 
6:    $\text{Miss}^{(t)} \leftarrow \mathcal{A} \setminus \hat{\mathcal{A}}^{(t)}$ 
7:    $\text{Extra}^{(t)} \leftarrow \hat{\mathcal{A}}^{(t)} \setminus \mathcal{A}$ 
8:    $s^{(t)} \leftarrow E(|M^{(t)}|, |\hat{\mathcal{A}}^{(t)}|, |\mathcal{A}|)$  ▷ Evaluation
9:   if  $s^{(t)} > \tau^*$  then ▷ Save best & update patience
10:     $\mathcal{Q}^* \leftarrow \hat{\mathcal{Q}}^{(t)}$ 
11:     $\tau^* \leftarrow s^{(t)}$ 
12:     $p \leftarrow 0$ 
13:   else
14:     $p \leftarrow p + 1$ 
15:   end if
16:   if  $\tau^* \geq \tau$  or  $p \geq m$  then ▷ Early exit
17:     break
18:   end if
   // Feedback signal creation
19:    $\Delta^{(t)} \leftarrow (M^{(t)}, \text{Miss}^{(t)}, \text{Extra}^{(t)}, s^{(t)}, \hat{\mathcal{Q}}^{(t)})$ 
   // Question Refinement
20:    $\mathcal{Q}^{(t+1)} \leftarrow \text{REFINE}(\hat{\mathcal{Q}}^{(t)}, \Delta^{(t)})$ 
21: end for
22: return  $\mathcal{Q}^*$ 

```

extraction performance:

$$\mathcal{Q}^* = \arg \max_{\mathcal{Q}} \text{F1}(\hat{\mathcal{A}}(\mathcal{D}, r, \mathcal{Q}), \mathcal{A}) \quad (1)$$

where $\hat{\mathcal{A}}(\mathcal{D}, r, \mathcal{Q})$ denotes the arguments extracted by a prediction model (PM) conditioned on $\mathcal{D}$, r , and $\mathcal{Q}$. We optimize $\mathcal{Q}$ via an iterative feedback loop (Algorithm 1) whose main components are:

Initialization. At $t=0$, the question generator (QG) model produces an initial list of questions in a zero-shot manner conditioned on r , $\mathcal{D}$, and $\mathcal{A}$.

Leakage Check. Since QG models can see $\mathcal{A}$, they could potentially leak ground-truth arguments ($a_i \in \mathcal{A}$) into the generated questions, directly or indirectly. To guard against this, we use an LM-based leakage check. At each iteration, $\text{LeakageCheck}(\mathcal{Q}^{(t)}, \mathcal{A})$ flags questions that potentially *leak* ground truth and rewrites them to remove the leak (e.g. asking a more generic question). We implement this module using GPT-OSS-120B. Appendix B.2 includes robustness analysis.

Extraction and Evaluation. Leakage-checked questions $\hat{\mathcal{Q}}^{(t)}$ are forwarded to the PM, which

outputs predicted arguments $\hat{\mathcal{A}}^{(t)}$. Predictions are evaluated against ground truth $\mathcal{A}$ using the evaluation framework (Section 4). This yields sets of matched, missing, and over-generated arguments, from which we compute precision, recall, and F1.

Feedback and Refinement. A structured feedback signal Δ at iteration t contains the scores, matched, missed, and over-generated arguments, and passed to a refiner LM. Based on the current questions $\mathcal{Q}^{(t)}$ and feedback $\Delta^{(t)}$, the refiner generates an updated set $\mathcal{Q}^{(t+1)}$. The refinement is instructed to add questions targeting missed arguments, refine or remove questions causing over-generation, and preserve effective questions.

Convergence. We track the best question set $\mathcal{Q}^*$ across iterations by F1 score. The loop terminates when (i) $\text{F1} \geq \text{target threshold}$, (ii) no improvement for p consecutive iterations (patience), or (iii) the maximum iteration count T is reached. The output is always $\mathcal{Q}^*$ (best, not necessarily final since the F1 score is not guaranteed to be monotonic).

3.3 Learning to Generate Questions

FEEDQ produces gold questions and, in doing so, yields a dataset of $(\mathcal{D}, r, \mathcal{Q}^*)$ triples. We use these triples to supervise the fine-tuning of Qwen3-4B and Qwen3-8B with LoRA. The resulting model learns to generate document- and role-conditioned questions without access to ground-truth, making LOQ applicable to unseen documents at test time.

3.4 Prediction

At inference time, the fine-tuned QG model generates questions conditioned on the document and target roles. A prediction model then uses these questions to extract arguments from the document. We evaluate five prediction models spanning both open-weight and proprietary LLMs: Qwen3-4B, Qwen3-8B, GPT-OSS-120B, GPT-5-Mini, and Gemini-3.1-Pro. Each PM is paired with every QG, yielding a complete comparison across both axes. Details in Appendix B.1, full prompts in Appendix C.

4 Datasets and Evaluation

We evaluate on four clinical and biomedical IE datasets spanning different text types and annotation granularities: **CaseReportBench** (Zhang et al., 2025), comprising full-length clinical case reports (avg. 582 words); **PHEE** (Sun et al., 2022), consisting of short pharmacovigilance texts (avg. 20

	Qw3 4B	Qw3 8B	OSS 120B	Gpt 5 mini	Gem- ini 3.1	Mean	Qw3 4B	Qw3 8B	OSS 120B	Gpt 5 mini	Gem- ini 3.1	Mean
Approach	CaseReportBench						PHEE					
No-Question	55.6	54.9	70.1	65.2	70.6	63.3	62.8	67.8	66.2	66.4	83.3	69.3
Performance with Knowledge Questions												
Knowledge-Q	62.4	61.6	75.3	71.0	78.3	69.7	76.5	79.5	75.7	71.8	83.4	77.4
CoT-Q	62.8	57.5	73.9	67.9	77.2	67.9	69.2	73.5	76.2	78.6	84.2	76.3
Performance with Dynamic Questions generated with GPT-OSS-120B												
Contextual-Q	54.3	50.6	57.9	52.5	57.7	54.6	49.3	47.5	51.0	52.9	79.5	56.0
Zs-FEEDQ	76.9	72.1	92.7	88.7	91.9	84.5	82.5	83.5	86.3	84.4	89.4	85.2
Opt-FEEDQ	79.2	73.4	96.9	92.4	94.0	87.2	82.3	84.8	94.3	89.8	90.6	88.4
	DiscourseEE						MACCROBAT					
No-Question	46.6	46.1	47.2	44.9	52.2	47.4	30.3	33.2	37.1	37.1	52.0	37.9
Performance with Knowledge Questions												
Knowledge-Q	53.0	53.8	60.5	55.6	57.8	56.1	36.6	38.3	42.1	42.5	58.1	43.5
CoT-Q	54.7	55.7	61.9	53.6	60.1	57.2	36.9	38.6	42.5	40.0	57.1	43.0
Performance with Dynamic Questions generated with GPT-OSS-120B												
Contextual-Q	42.2	41.3	43.5	41.0	40.3	41.7	30.5	31.3	30.5	29.3	38.3	32.0
Zs-FEEDQ	60.0	61.2	72.2	61.2	67.3	64.4	49.4	56.3	74.9	64.1	83.6	65.6
Opt-FEEDQ	65.5	64.8	82.5	64.6	69.0	69.3	57.3	66.5	91.1	74.6	91.3	76.2

Table 1: Impact of question quality on dev set extraction performance across four datasets and five prediction models. For each column, only the questions change, isolating the effect of asking the right questions. Zs-FEEDQ are questions generated in one zero-shot pass and Opt-FEEDQ are questions iteratively refined with FEEDQ.

words); **DiscourseEE** (Sharif et al., 2024), drawn from informal online health forums; and **MACCROBAT** (Ma et al., 2023), containing PubMed case report snippets. All datasets are transformed into the unified format described in Section 3.1: triplets of (document, role, list of ground-truth arguments). Data statistics are given in Table 8.

Each dataset is divided into training, development, and test splits. The development set is used to design and internally validate FEEDQ. The training set provides the samples on which FEEDQ produces gold questions for fine-tuning. **The test set is held out for final evaluation on unseen data.** More details are in Appendix A. We report F1 for all approaches on test set. Exact-match and relaxed-match metrics are known to underestimate LLM performance, as semantically correct predictions with surface-form differences are penalized (Lu et al., 2025; Fane et al., 2025). We adopt the hierarchical approach of Sharif et al. (2025), which combines exact, relaxed, and LLM-as-judge to evaluate argument correctness (details in Appendix B.5).

5 Importance of Better Questions

To first **isolate the effect of question quality** on extraction, we compare approaches that vary only

how questions are formed while holding the prediction model fixed (Table 1). In No-Question, we prompt the prediction model with (document, role) and instruct it to generate arguments without questions. Performance degrades across all datasets, establishing that questions are necessary. Knowledge-Q and CoT-Q are *knowledge questions*: defined once per role and applied uniformly across all documents (Du and Cardie, 2020; Hsu et al., 2022). These are directly taken from expert questions released with the dataset or minimally adapted to QA format from their expert annotation guidelines (details: Appendix B.3). These are a strong human-written question baseline. Contextual-Q generates *dynamic questions* per document–role pair *without* access to ground truth, a realistic deployment setting (Hong and Liu, 2024). Zs-FeedQ and Optimized-FeedQ also generate dynamic questions, but initially *with* access to ground-truth arguments: the former in a single zero-shot pass, the latter through iterative refinement via FEEDQ. Ground-truth access **establishes an upper bound on what is achievable when questions are well-targeted**, before asking whether that quality can be reproduced without it. For experiments, we use the prediction models discussed in Section 3.4.

One key observation is that Contextual-Q degrades below No-Question baseline. This raises the question of why document-condition questions hurt the extraction performance. To investigate this, we did a precision-recall analysis, shown in Table 2. We find that **unoptimized Contextual Questions lead the prediction models to generate many extra candidate arguments, inflating false positives.**

Dataset	Approach	P	R	F1
CaseReportB	No-Question	54.5	77.5	63.3
	Contextual-Q	40.2	85.8	54.6
PHEE	No-Question	63.6	76.9	69.4
	Contextual-Q	44.6	78.6	56.0
DiscourseEE	No-Question	37.4	65.7	47.4
	Contextual-Q	28.8	76.3	41.7
MACCROBAT	No-Question	29.1	54.8	38.0
	Contextual-Q	22.1	57.9	32.0

Table 2: Overall precision, recall, and F1 analysis of the No-Question and Contextual-Q approaches.

Both No-Question and Contextual-Q condition on the document and the target role; Contextual-Q additionally provides questions generated on the fly without ground-truth access. **It generates 5.27 questions per document–role pair on average which causes the trade-off above.** Across every dataset it raises recall (e.g., DiscourseEE 65.7 → 76.3) but lowers precision by large margin (37.4 → 28.8). In contrast, No-question produces fewer arguments, so its recall is lower. However, large prediction models can still recover many common arguments directly from the document and role specification, resulting in substantially higher precision and, ultimately, higher F1.

5.1 Discovering the Right Questions

We select the best questions using FEEDQ. Selecting a question generator for FEEDQ requires balancing quality and cost. We evaluate five LLMs as zero-shot question generators (Qwen3-4B, Qwen3-8B, GPT-OSS-120B, GPT-5.2, and Gemini-3.1-Pro), the latter two with high reasoning. For each, we compute the F1 score averaged across all prediction models. Table 3 summarizes the results; per-model breakdown is illustrated in Figure 5.

Larger models outperform the Qwen3 baselines by a wide margin (≈ 72 – 76 vs. ≈ 54 F1). GPT-5.2 performs best overall (76.1), but GPT-OSS-120B is close (74.9) while being over $40\times$ cheaper per token at the time of writing. We therefore use GPT-

QG	CRB	PHEE	DEE	MBAT	Mean
Qwen3-4B	60.8	66.8	48.3	43.3	54.8
Qwen3-8B	60.8	64.1	47.8	40.7	53.4
OSS-120B	84.5	85.2	64.4	65.6	74.9
GPT-5.2	85.1	85.9	65.8	67.8	76.1
Gemini-3.1	79.8	85.1	63.5	59.8	72.1

Table 3: Zero-shot question generation quality across five LLMs, measured by F1 averaged over prediction models. Per-model scores shown in Figure 5. CRB, DEE, and MBAT indicate CaseReportBench, DiscourseEE, and MACCROBAT datasets, respectively.

OSS-120B as the QG model. Its open weights additionally allow local deployment and use with private data when needed. For scalability, we also use GPT-OSS-120B as the prediction and refinement model, though any sufficiently capable model could be substituted.

5.2 Impact of Question Improvement

Better questions would yield only marginal gains if the bottleneck in LLM-based extraction were model capability. Table 1 results suggest otherwise. Optimized-FeedQ outperforms the strongest baseline (Knowledge-Q) by **18.6 F1** averaged across four datasets and all prediction models. Gains hold on every dataset–model combination, ranging from 11.0 (PHEE) to 32.7 (MACCROBAT).

Zero-shot FeedQ already exceeds Knowledge-Q by 13.2 F1 without iterative refinement, indicating that much of the gain comes from generating questions informed by what information the document actually contains for a given role. FeedQ optimization adds a further 5.4 F1, with the largest gain on DiscourseEE (+4.9) and MACCROBAT (+10.6), where arguments are most frequently implicit or distributed. Neither alternative questioning strategy helps: Contextual-Q falls 15.6 F1 below Knowledge-Q despite conditioning on the document, and CoT-Q shows no benefit either (61.1 vs. 61.7). **Document conditioning without grounding produces vague queries and reasoning depth does not compensate for a poorly targeted question.**

5.3 Analysis of Optimized Questions

Figure 3 and 4 provide justification of two core design choices in our pipeline. First, iterative refinement is not redundant: while a single FEEDQ pass suffices for the majority of roles in PHEE (87%) and MACCROBAT (75%), nearly half of DiscourseEE roles (47%) and a quarter of CaseReport-

	Qwen3 4B	Qwen3 8B	OSS 120B	Gpt-5 mini	Gemini 3.1-pro	Qwen3 4B	Qwen3 8B	OSS 120B	Gpt-5 mini	Gemini 3.1-pro
Approach	CaseReportBench					PHEE				
Performance with Knowledge Questions										
Knowledge-Q	60.29	60.17	73.40	69.03	78.17	76.92	80.02	74.08	74.05	81.37
CoT-Q	57.98	58.24	73.92	66.33	76.32	67.98	73.67	77.50	76.41	80.53
Performance with fine-tuned and non fine-tuned QG models										
Qwen3-4B	63.18	59.26	52.28	45.73	66.14	64.71	63.54	62.30	58.75	79.89
Qwen3-8B	63.26	60.51	56.67	49.71	67.33	62.10	63.26	58.58	55.42	79.11
Qwen3-4B ^{ft*}	62.56	64.09	71.63	67.86	71.32	78.48	81.52	83.61	80.30	84.96
Qwen3-8B ^{ft*}	63.85	64.59	70.88	69.16	69.98	78.07	80.66	84.52	82.74	84.25
Strong QG Baselines (non fine-tuned)										
Gpt-OSS-120B	53.76	49.45	56.31	51.14	55.88	48.57	44.87	49.40	49.81	77.35
Gpt-5.2	60.46	58.07	69.36	64.13	65.63	62.59	61.39	71.87	71.44	80.33
Gemini-3.1	62.01	59.39	70.21	67.04	71.26	79.39	80.68	80.98	80.13	81.89
	DiscourseEE					MACCROBAT				
Performance with Knowledge Questions										
Knowledge-Q	50.55	53.68	60.36	55.15	58.13	37.55	39.11	43.70	41.36	58.74
CoT-Q	50.44	55.13	61.01	52.71	59.62	37.41	41.75	43.60	41.60	58.51
Performance with fine-tuned and non fine-tuned QG models										
Qwen3-4B	50.11	48.75	47.13	44.96	47.64	38.02	41.50	41.07	38.72	48.43
Qwen3-8B	47.91	48.90	46.59	42.66	44.36	38.71	40.28	40.12	39.07	47.57
Qwen3-4B ^{ft*}	49.75	50.16	54.12	54.01	51.46	45.37	46.37	51.39	48.62	56.84
Qwen3-8B ^{ft*}	47.93	50.57	56.74	54.47	52.71	41.81	44.96	50.96	47.53	56.74
Strong QG Baselines (non fine-tuned)										
Gpt-OSS-120B	41.66	40.52	44.60	41.59	38.83	32.49	32.51	33.31	32.90	39.18
Gpt-5.2	48.89	43.95	50.55	46.65	41.04	39.57	39.88	40.50	39.30	42.91
Gemini-3.1	51.39	49.65	53.18	50.84	49.52	37.13	40.59	43.25	41.49	52.13

Table 4: Test-set performance comparison across question generation approaches. Knowledge-Q and CoT-Q are human-written questions. All other approaches generate questions dynamically, conditioned only on the document and role (no ground-truth access). Qwen3^{ft} rows report the best score after fine-tuning (see Table 7 for the full breakdown). Bold indicates the fine-tuned model beats all approaches for that dataset; green and light green mark the first and second highest scores overall.



Figure 3: Distribution of the FEEDQ iteration at which the best-scoring question set is found in the dev set.

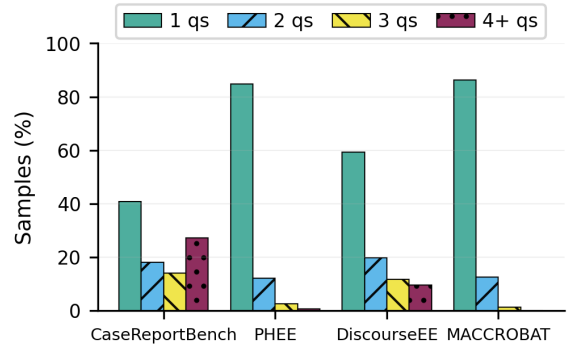


Figure 4: Distribution of the percentage of optimized questions generated per role in the development set.

Bench roles require two or more iterations before reaching their best-scoring question set (Figure 3). Second, a single question per role is frequently insufficient. CaseReportBench assigns only 41% of

roles a single optimized question; the remaining 59% receive two or more, with 27% requiring four or more (Figure 4). DiscourseEE follows a similar pattern. This confirms that roles in structurally

rich schemas often demand multiple targeted questions to adequately decompose the extraction target. For PHEE and MACCROBAT, single well-phrased questions work well enough in the majority of cases. In Appendix D, we add supplementary evidence showing *how* the questions change using auto-interpretability methods.

6 Learning the Right Questions

Section 5 established that better questions improve extraction. The practical question is whether a model can learn to generate effective questions for *unseen* documents, i.e. without access to ground-truth arguments. We test this by fine-tuning Qwen3-4B and Qwen3-8B models on the questions produced by FEEDQ. Specifically, we run FEEDQ on the training split of each dataset to generate (document, role, optimized question-set) triples, producing **12,825 triples** in total. We then perform supervised fine-tuning with LoRA to train each model to map a document and role to a question (Section 3.3). We construct three training configurations: a *balanced mix* that caps each dataset at a comparable sample count ($\approx 6K$ total), an *all-available* mix that uses every training sample ($\approx 12K$ total), and an *in-domain* mix that trains on only the target dataset’s triples. We report the best of the three per model and provide all comparisons in Table 7. More fine-tuning details are in Appendix B.4.

Approach	CRB	PHEE	DEE	MBAT	Mean
<i>Knowledge Questions</i>					
Knowledge-Q	68.2	77.3	55.6	44.1	61.3
CoT-Q	66.6	75.2	55.8	44.6	60.5
<i>Fine-tuned and Non Fine-tuned QG Models</i>					
Qwen3-4B	57.3	65.8	47.7	41.5	53.1
Qwen3-8B	59.5	63.7	46.1	41.1	52.6
Qwen3-4B ^{ft-a}	67.5	80.7	51.9	49.0	62.3
Qwen3-4B ^{ft*}	67.5	81.8	51.9	49.7	62.7
Qwen3-8B ^{ft-a}	64.7	81.6	51.5	47.2	61.3
Qwen3-8B ^{ft*}	67.7	82.0	52.5	48.4	62.7
<i>Strong QG Baselines (Non Fine-tuned)</i>					
OSS-120B	53.3	54.0	41.4	34.1	45.7
Gpt-5.2	63.5	69.5	46.2	40.4	54.9
Gemini-3.1	66.0	80.6	50.9	42.9	60.1

Table 5: Test-set F1 averaged across five prediction models. Fine-tuned Qwen3 models (ft) are trained on optimized questions from FEEDQ; (ft* is best configuration per-model, ft-a is mixture-trained; full breakdown in Table 7). CRB, DEE, and MBAT indicate CaseReportBench, DiscourseEE, and MACCROBAT, respectively.

We compare against three groups of approaches.

First, the Knowledge-Q and CoT-Q which require human written questions (Section 5). Second, non-fine-tuned Qwen3-4B and Qwen3-8B generating questions for each document–role pair via contextual prompting (conditioned on document and role, no ground truth), isolating the effect of fine-tuning within the same models. Third, three strong non-fine-tuned QG models (GPT-OSS-120B, GPT-5.2, and Gemini-3.1-Pro) using the same contextual prompt, establishing what scale alone can achieve. **All results are reported on the held-out test set.** Table 4 gives per-model scores and Table 5 averages across prediction models.

6.1 Tuned models generate effective questions

Fine-tuning on FEEDQ-generated data substantially improves question generation quality. After tuning, both Qwen3 models reach 62.7 mean F1, up from ≈ 53 without fine-tuning (Table 5). More importantly, the fine-tuned models outperform all non-fine-tuned strong baselines, including GPT-OSS-120B, GPT-5.2, and Gemini-3.1. The largest gains appear on PHEE (≈ 16 – 18) and MACCROBAT (≈ 7 – 8), but the overall improvement is not dataset-specific. These results suggest that effective question generation depends largely on implementing the right questioning behavior.

Fine-tuned models beat Knowledge-Q on average (62.7 vs. 61.3). On PHEE and MACCROBAT, fine-tuned models outperform Knowledge-Q by 4.5–4.7 and 4.3–5.6 F1 respectively. On CaseReportBench, Knowledge-Q retains a marginal edge (under 1 F1), while on DiscourseEE the gap is larger (3–4 F1). This aligns with training data availability: PHEE and MACCROBAT each contribute 5,000 training samples, while CaseReportBench provides only 620 and DiscourseEE 2,215. With limited supervision, the fine-tuned models cannot fully learn the question patterns needed for these datasets. We provide systematic data scaling experiments in Appendix B.6.

6.2 Out-of-domain Experiments

We run additional zero-shot out-of-domain experiments on a subset of 500 samples on datasets: DocEE (Tong et al., 2022), which has long, argument-dense tasks; GENEVA (Parekh et al., 2023) which has brief, simple examples; and MUC4 (Sundheim, 1992), news documents. Overall, we find that fine-tuning benefits DocEE (the more long and complex samples), MUC4 but not GENEVA, which annotates multiple coreferent arguments per role and is

better served by the broader question sets of the base model. Precision-recall comparison indicates that the fine-tuned model produces role-targeted questions that improve precision across all datasets. Detailed analysis is in Appendix B.7.

Dataset	Qwen3-4B			Qwen3-8B		
	P	R	F1	P	R	F1
<i>Base</i>						
DocEE	33.35	82.88	46.53	35.58	83.82	49.20
GENEVA	48.33	77.51	59.13	47.38	77.66	58.30
MUC4	45.98	80.20	57.45	47.21	80.14	58.56
<i>Fine-tuned (ft*)</i>						
DocEE	49.18	63.63	55.28	48.38	68.65	56.65
GENEVA	49.73	56.43	52.82	49.66	56.65	52.88
MUC4	53.82	61.86	57.30	54.34	67.24	59.90

Table 6: Out-of-domain results on DocEE, GENEVA, and MUC4, averaged across all evaluator models (ft* is best configuration per model; full breakdown in Table 11). The highest F1 per dataset is bolded.

7 Conclusion

Extraction is a central ambition of information systems: how to find the right fact or passage when needed. However, documents are not databases; the information is not neatly organized awaiting extraction, and complex extractions involve representations of the problem and the data available to solve it. The query is inevitably part of this representation, and choosing it poorly can thus put an extractor on the wrong trail and render the necessary information invisible. In this paper, we have offered evidence of this problem and derived a scalable and automated solution. We invite the information extraction research community to draw on our methods and data to build better forms of information access for modern language technologies.

Limitations

One limitation is that FEEDQ uses GPT-OSS-120B as the sole model for question generation, refinement, leakage checking, and prediction during optimization. The optimized questions transfer effectively across all five prediction models at test time (Tables 4–5), indicating that the learned patterns generalize beyond the optimizer. Exploring whether varying the optimizer or ensembling questions from multiple models yields further gains could yield valuable additional design knowledge.

Separately, all four primary benchmarks in this work are clinical and biomedical, where arguments

are distributed across narrative sections and expressed in domain-specific terms. These properties that make targeted questioning particularly impactful. To further assess generalizability, we experimented with three out-of-domain non-clinical datasets. Results indicate that LOQ remains effective in these settings. Extending LOQ to other extraction domains such as news, law, or finance would demonstrate broader applicability.

Reproducibility

The datasets, models, and implementation details are provided in Appendices A, B, and C. Our code and additional resources are available at <https://omar-sharif.github.io/LoQ>.

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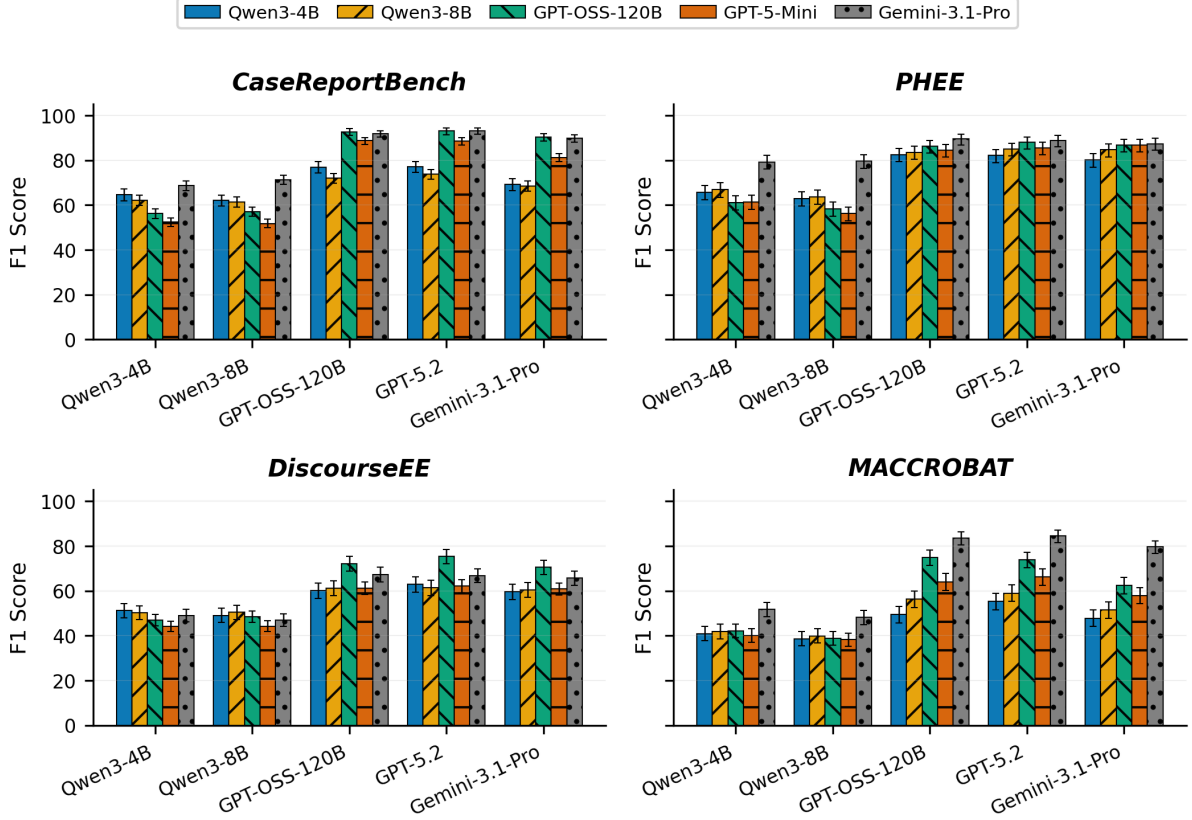


Figure 5: Per-model breakdown of zero-shot question generation quality. Each group of bars shows F1 scores for a single QG model evaluated with all five prediction models on each dataset. Larger QG models (GPT-OSS-120B, GPT-5.2, Gemini-3.1-Pro) consistently outperforms the smaller Qwen variants across all datasets and prediction models. Averages are reported in Table 3.

	Qwen3 4B	Qwen3 8B	OSS 120B	Gpt-5 mini	Gemini 3.1-pro	Qwen3 4B	Qwen3 8B	OSS 120B	Gpt-5 mini	Gemini 3.1-pro
QG Model	CaseReportBench					PHEE				
Qwen3-4B ^{ft-b}	60.64	59.43	60.68	62.17	60.95	75.76	78.60	79.89	79.69	82.20
Qwen3-4B ^{ft-a}	62.56	64.09	71.63	67.86	71.32	78.22	81.52	81.00	78.97	83.92
Qwen3-4B ^{ft-in}	58.98	60.01	63.15	61.59	60.77	78.48	80.24	83.61	80.30	84.96
Qwen3-8B ^{ft-b}	62.98	61.59	69.40	66.07	67.89	77.42	77.38	78.88	77.18	81.35
Qwen3-8B ^{ft-a}	60.89	61.03	69.22	65.64	66.83	78.07	80.66	82.36	82.74	84.25
Qwen3-8B ^{ft-in}	63.85	64.59	70.88	69.16	69.98	77.86	79.99	84.52	80.03	83.70
	DiscourseEE					MACCROBAT				
Qwen3-4B ^{ft-b}	45.27	47.61	49.36	50.38	46.22	40.61	45.22	49.52	45.45	53.18
Qwen3-4B ^{ft-a}	49.75	50.16	54.12	54.01	51.46	41.71	46.37	51.39	48.62	56.84
Qwen3-4B ^{ft-in}	47.69	48.57	51.35	51.88	45.96	45.37	45.97	49.20	46.54	55.34
Qwen3-8B ^{ft-b}	47.11	49.20	56.74	54.47	52.71	41.81	44.96	50.64	46.24	56.22
Qwen3-8B ^{ft-a}	47.72	50.57	55.66	53.21	50.21	41.02	43.25	50.96	47.53	53.36
Qwen3-8B ^{ft-in}	47.93	47.63	50.86	53.37	48.73	40.95	44.69	50.90	45.45	56.74

Table 7: Impact of fine-tuning data composition on QG model performance. Each row fine-tunes the QG model (Qwen3-4B or Qwen3-8B) on a different subset of FEEDQ-optimized questions: *ft-b* uses a balanced mix sampling equally across datasets, *ft-a* uses all available optimized questions regardless of source distribution, and *ft-in* uses only in-domain questions matching the target dataset. The best F1 score of the ablation is reported in Table 4.

A Datasets

All datasets are converted to the unified format described in Section 3.1: each sample consists of a document, a role, and its ground-truth arguments. We create development and test splits of 500 samples per dataset, except CaseReportBench where the smaller original dataset limits both to 350. Table 8 reports detailed statistics. Descriptions of each dataset provided in the following.

CaseReportBench (Zhang et al., 2025). is a dense information extraction dataset from clinical case reports on rare diseases. Each case report is annotated by clinical experts into different categories such as laboratory findings, diagnosis, treatment, and outcomes. This dataset targets the extraction of structured information from long, unstructured clinical narratives requiring reasoning over entire documents. We source the dataset from author’s HuggingFace repository <https://huggingface.com/CaseReportBench>. We map it into our unified format by treating each category as a role with its corresponding argument strings.

PHEE (Sun et al., 2022). is an argument extraction dataset sourced for the pharmacovigilance (i.e. drug safety) domain. The dataset contains 14 different roles covering aspects such as drug, dosage, route, frequency, duration, disorder, and patient demographics (age, gender, race). We obtain the dataset from official repo <https://github.com/ZhaoyueSun/PHEE>. The original dataset includes both trigger and argument span annotations at the sentence level. Following our formulation, we only take the argument strings and combine multiple arguments under the same role into a single argument list.

DiscourseEE (Sharif et al., 2024). A dataset annotated from online health forum about information seeking posts on opioid use disorder. It features 34 unique roles across with all arguments annotated as free-text strings. Notably, 51.2% of the arguments are implicit (inferable from context but not directly stated) making this dataset uniquely challenging for argument extraction. We sourced the data from official repository <https://github.com/omarsharif03/DiscourseEE> and transform it into our format following author’s provided code.

MACCROBAT (Ma et al., 2023). is a clinical event extraction dataset repurposed from expert-annotated PubMed case reports originally introduced by Caufield et al. (2019). It contains 22 argument roles inferred from entity-event modification relations. The dataset is notably dense, averaging 3.21 events per sentence. We collect the repurposed dataset from official GitHub repository <https://github.com/DICE>.

B Implementation Details

B.1 Models

We evaluate across three model families *Qwen*, *GPT*, and *Gemini* — spanning open-source and proprietary LLMs at different scales. For the Qwen (Team, 2025), we use the Qwen3-4B and Qwen3-8B in non-thinking mode. Beyond prediction, we fine-tune both models with LoRA for question generation (Section 3.3). We use three GPT¹ models: Gpt-OSS-120B serves as both a prediction and question generation model with identical hyperparameters; GPT-5-mini with medium reasoning effort serves as a prediction model; and GPT-5.2 with high reasoning effort serves as a strong question generation model. Finally, we evaluate Gemini-3.1-Pro² with high reasoning for both question generation and prediction.

B.2 FEEDQ Parameters and Leakage Analysis

We run FEEDQ for up to 5 iterations with a patience of 3 and a target score of 1.0 (i.e., full extraction of all ground-truth arguments). Figure 3 confirms that 5 iterations are sufficient: across all four datasets, the best-scoring question set is found within this budget, though the number of iterations required varies by dataset complexity.

Leakage check. Because FEEDQ optimizes questions against ground-truth arguments, the optimizer could embed answer fragments in the question text. We guard against this with two measures. First, the question refinement prompt (Appendix C.3) explicitly instructs the model not to include answer-related information. Second, a standalone leakage checker inspects every generated question set and removes any question containing ground-truth content. Here, we analyze the checker’s behavior systematically on the development set. Table 9 reports

¹<https://openai.com/index/introducing-gpt-5-2/>

²<https://deepmind.google/models/gemini/pro/>

Dataset	Data Splits				General Statistics			
	Train	Dev	Test	Total	#Roles	Avg. doc length	Avg. argument length	Domain
CaseReportBench	620	350	350	1320	18	581.7	20.71	Clinical Reports
PHEE	5000	500	500	6000	14	19.7	2.45	Pharmacovigilance texts
DiscourseEE	2215	500	500	3215	34	120.8	3.04	Online Health Forums
MACCROBAT	5000	500	500	6000	22	24.2	1.73	PubMed Reports

Table 8: Summary statistics of the four datasets used in our experiments, including split sizes and general statistics.

the proportion of iterations in which the leakage checker flags at least one generated question as containing ground-truth content *before* filtering. Leakage occurs substantially across all four datasets, affecting 42–55% of iterations and 45–65% of samples. **The checker removes these questions before they enter the QG training set**, preventing downstream models learning from a flawed signal.

To validate the checker, we manually inspect 600 filtered question sets (150 per dataset) for missed leakage. **The false negative rate is 0.83% (5/600), confirming that the filter reliably catches answer-leaking questions.** Beyond filtering, our test-time setup eliminates ground-truth access entirely: fine-tuned QG models generate questions for unseen documents with no oracle, and outperform strong QG models and all knowledge-driven baselines on average (Tables 4–5). If the gains were driven by implicit encoding of target information during FEEDQ optimization, we would expect this transfer to degrade. Instead, it improves over the base model. The gain also holds under domain shift: on DocEE (out-of-domain), fine-tuning improves Qwen3-8B by +7.4 F1 (Table 6). Implicit answer encoding tied to the optimization documents or semantic leakage would not transfer to a new domain with different argument distributions. Taken together, these analyses indicate minimal leakage.

Dataset	Iters Leaked (%)		Samples w/ Leak (%)	
CRB	280 / 505	(55.4)	219 / 350	(62.6)
PHEE	272 / 644	(42.2)	223 / 500	(44.6)
DEE	516 / 1104	(46.7)	323 / 500	(64.6)
MBAT	353 / 816	(43.3)	228 / 500	(45.6)

Table 9: Ground-truth leakage in FEEDQ-generated questions on the dev set, measured before filtering. *Iter. Leaked* counts optimization iterations that produced at least one question containing a verbatim argument; *Samples w/ Leak* counts unique samples affected.

B.3 Knowledge Questions Source

We take these human-written questions directly from materials released by the dataset authors, or minimally adapt them to QA form from the expert annotation guidelines. For DiscourseEE, we use the expert questions released with the original paper (Sharif et al., 2024) verbatim. For CaseReportBench (Zhang et al., 2025), we adapt the role definitions in the Expert Annotation Guidelines released in the dataset’s GitHub repository.³ For PHEE (Sun et al., 2022) and MACCROBAT (Ma et al., 2023; Caufield et al., 2019), we adapt the role-level definitions provided in the papers. Therefore, this constitutes a strong human baseline questions.

B.4 Fine-tuning

We fine-tune Qwen3-4B and Qwen3-8B using LoRA (rank 32) with Tinker⁴. All models are trained for 2 epochs with a learning rate of $3.7e^{-4}$ and maximum sequence length of 4,096. We construct three training mixes (Table 10): a balanced mix (6,200 triples) that caps each dataset at a comparable count, all-available mix (12,825) triples that uses all the samples, and *in-domain* mix that trains on only the target dataset’s triples. In all cases, 20% is reserved for validation to select the best checkpoint. We report high score per model in Table 4 and provide the full breakdown in Table 7.

Dataset	Balanced (ft-b)	All (ft-a)
CaseReportBench	620	620
PHEE	1,860	5,000
DiscourseEE	1,860	2,215
MACCROBAT	1,860	5,000
Total	6,200	12,825

Table 10: Number of $(\mathcal{D}, r, Q^*)$ triples per dataset in each fine-tuning mix. We reserve 20% for validation to select the best checkpoint and use the remaining 80% for training.

³<https://tinyurl.com/vvrautrsr>

⁴<https://tinker-docs.thinkingmachines.ai/supervised-learning>

B.5 Evaluation

We follow REGen (Sharif et al., 2025), a three-level hierarchical evaluation framework for generative argument matching. A predicted argument is considered correct if it matches a ground-truth argument at any of three levels: exact, similarity-based relaxed, or LLM-as-judge based complex match. Precision/recall/F1 are then measured the standard way. At relaxed match level, two arguments are considered similar if their SBERT embedding similarity (Reimers and Gurevych, 2019) exceeds 0.85. For complex matching, we use GPT-OSS-120B as a judge model with the prompt below.

Prompt for judge-based matching

Determine if two arguments refer to the same core entity, value, or claim for the given role. Match if they convey the same core meaning, even with minor differences in detail, formatting, or specificity.

Context: {context}

Role: {role}

Argument 1: {x}

Argument 2: {y}

Answer "yes" or "no" only.

B.6 Impact of Data Scaling

We measure how QG fine-tuning scales with training data. We fine-tuned Qwen3-8B QG model on each dataset’s randomly sampled triples at 10%, 20%, 30%, and 100% fractions, and evaluated with GPT-OSS-120B. Figure 6 reports the results.

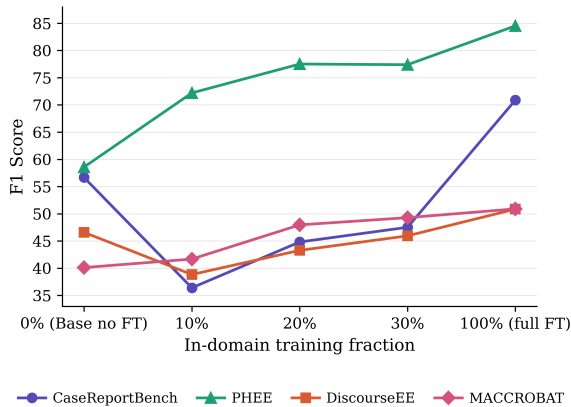


Figure 6: Effect of dataset scaling on QG fine-tuning. A separate Qwen3-8B QG model is fine-tuned on 10/20/30/100% of each dataset’s triples.

PHEE and MACCROBAT are larger, with 5,000 triples each. Even 10% of the data improves over the base model, and performance rises monotonically as more data is added. CaseReportBench and DiscourseEE behave differently. Both are small: 620 and 2,215 triples in total. At 10%, both drop below the base model. The drop is consistent with catastrophic forgetting from fine-tuning on a small in-domain sample. Performance recovers as more data is added and surpasses the base by 100%. Across all four datasets, more training data improves model performance, showing the clear impact of dataset scaling.

B.7 Out-of-domain Analysis

Fine-tuning improves Qwen3-8B by up to +7.4 F1 on DocEE. DocEE samples are long (often several thousand words), and the base model generates a large pool of loosely relevant questions. The fine-tuned model produces a small set of targeted questions. Fine-tuning generalizes the question-asking behavior beyond the training domains rather than memorizing question patterns specific to them. MUC4 shows the same pattern, with a smaller +1.3 F1 gain. Our approach therefore delivers significant in-domain gains while preserving performance under substantial out-of-domain shift.

DocEE example (full document: 948 words, 5,647 chars)

Excerpt: “The 1931 Hawke’s Bay earthquake [...] occurred in New Zealand at 10:47 am on 3 February, killing 256, injuring thousands [...] Nearly all buildings in the central areas of Napier and Hastings were levelled [...] the death toll included 161 people in Napier, 93 in Hastings, and two in Wairoa. Thousands more were injured, with over 400 hospitalised [...]”

Base model (30 questions, many off-target):

- How many people were killed in the 1931 Hawke’s Bay earthquake?
- What was the total number of injuries reported during the earthquake?

...

Fine-tuned model (4 questions, targeted to roles and context):

- What were the total number of deaths and injuries reported from the 1931 Hawke’s Bay earthquake?
- How many people were killed in each of the affected cities (Napier, Hastings, Wairoa)?

...

Why model fail on GENEVA: GENEVA’s annotation style frequently assigns multiple coreferent

	Qwen3 4B	Qwen3 8B	OSS 120B	Gpt-5 mini	Gemini 3.1-pro	Mean	Qwen3 4B	Qwen3 8B	OSS 120B	Gpt-5 mini	Gemini 3.1-pro	Mean
QG Model	DocEE						GENEVA					
Qwen3-4B	56.94	48.23	36.00	36.05	55.41	46.53	55.37	54.55	54.36	54.57	76.79	59.13
Qwen3-8B	58.26	50.44	41.22	40.82	55.28	49.20	54.07	51.26	53.93	53.46	78.76	58.30
Qwen3-4B ^{ft-b}	59.42	51.29	51.92	55.38	53.90	54.38	48.79	47.30	47.30	49.52	63.43	51.27
Qwen3-4B ^{ft-a}	57.98	51.63	55.01	57.80	53.97	55.28	48.18	52.21	48.87	52.86	61.96	52.82
Qwen3-8B ^{ft-b}	59.27	52.50	56.22	58.63	56.61	56.65	50.59	49.56	47.47	50.44	66.35	52.88
Qwen3-8B ^{ft-a}	54.04	50.04	51.96	56.63	55.46	53.63	49.11	49.51	47.01	50.21	64.18	52.00
QG Model	MUC4											
Qwen3-4B	64.08	58.64	47.08	51.08	66.36	57.45	-	-	-	-	-	-
Qwen3-8B	62.64	58.48	51.86	53.08	66.73	58.56	-	-	-	-	-	-
Qwen3-4B ^{ft-b}	56.04	55.18	52.80	61.40	61.07	57.30	-	-	-	-	-	-
Qwen3-4B ^{ft-a}	57.46	54.74	51.74	59.02	60.71	56.73	-	-	-	-	-	-
Qwen3-8B ^{ft-b}	59.47	56.02	57.46	63.35	63.22	59.90	-	-	-	-	-	-
Qwen3-8B ^{ft-a}	57.82	53.74	53.38	60.02	61.94	57.38	-	-	-	-	-	-

Table 11: Out-of-domain generalization on DocEE, GENEVA, and MUC4. Non-fine-tuned QG models (top) and fine-tuned variants (bottom). Models are fine-tuned with FEEDQ-optimized questions directly. We report f1 score with two variants (*ft-b*: balanced mix, *ft-a*: all available). Bold indicates the best f1 score in each column within a dataset.

or syntactically distinct arguments to the same role within a single sentence. Recovering all such arguments benefits from probing the role from several complementary angles. The broader question sets produced by the base model happen to provide this additional coverage, even though such breadth would introduce noise on datasets with cleaner annotations. The fine-tuned QG, by contrast, produces more focused, role-targeted questions, which improves precision but reduces overall F1 due to lower coverage.

GENEVA annotation examples

Example 1. *Excerpt*: “The best beaches have become home to the finest hotels, which supply almost everything needed for the perfect vacation.”

Role: supply: supplier

Args: {“the finest hotels”, “which”}

The noun phrase and its relative pronoun are both annotated as gold arguments within the same clause.

Example 2. *Excerpt*: “Many islands had already transferred their allegiance to the victors, who were led by Philip II of Macedon.”

Role: sending: recipient

Args: {“to the victors”, “Philip II of Macedon”}

Both the prepositional recipient and the named leader of that group are gold annotations.

Example 3. *Excerpt*: “... defines terrorism as: ‘The sub-state application of violence or threatened violence...’”

Role: using: agent

Gold: {“sub-state”}

The descriptor is annotated as the agent, requiring a question that targets the modifier rather than the entity.

C Prompts

C.1 Leakage Check

You are a quality control reviewer for an information extraction system. Your task is to check whether any of the generated questions contain **information leakage**—meaning the questions directly reveal, embed, or strongly hint at specific ground truth arguments that they are supposed to help extract.

Why this matters: The questions should guide an extraction model to find relevant information in a document, NOT hand the answers to it. A leaked question makes extraction trivially easy, but produces questions that won’t generalize to new, unseen documents.

Leakage (MUST be rewritten)

A question has **strong leakage** if it:

1. **Directly contains** a ground truth argument verbatim or near-verbatim (e.g., asking “Is the patient’s hemoglobin 7.2 g/dL?” when “hemoglobin 7.2 g/dL” is a ground truth argument)
2. **Embeds specific values, dosage, or measurements** from the ground truth (e.g., “Was the patient started on levetiracetam 500mg twice daily?” when that exact medication regimen is a ground truth argument)

Acceptable / Minor Overlap (keep unchanged)

A question is **acceptable** and should NOT be rewritten if it:

- Asks about a general clinical category or organ system (e.g., “What hematological abnormalities were identified?” is fine even if a specific hematological

finding is in the ground truth)

- Uses standard medical terminology that naturally overlaps with ground truth terms (e.g., mentioning “neurological” when neurological findings exist in ground truth)
- Asks open-ended questions that probe a relevant clinical domain without specifying exact answers
- References broad clinical concepts like symptom categories, diagnostic approaches, or treatment classes

Input

Role: {role}

Current Questions: {current_questions}

Ground Truth Arguments: {gt_arguments}

Instructions

1. Review each question against all ground truth arguments for strong leakage only
2. For any question with leakage, rewrite it to be more general while still targeting the same information category
3. For all other questions, keep them exactly unchanged

Respond ONLY with valid JSON in this exact format:

```
{
  "is_leaked": "yes/no",
  "final_questions": [
    "rewritten or original question 1",
    "rewritten or original question 2"
  ]
}
```

Set "is_leaked" to "yes" only if at least one question had strong leakage and was rewritten. Otherwise, set it to "no". Do not include any text outside the JSON.

C.2 Question Generation

You are a helpful assistant that generates clear and distinct questions to best extract the ground truth arguments for a given role.

Role: {role}

Document: {document}

Ground truth arguments: {gt_arguments}

Instructions:

- Each question should aim to extract one or more ground truth arguments.
- If a single question can logically extract multiple arguments, group them in one question.
- Avoid redundancy—do not create multiple questions that ask for the same information.
- Avoid yes/no questions or vague phrasing.
- Keep questions concise so the extraction model does not get confused and extract irrelevant arguments.

Examples:

```
{
  "subject": "Individual or patient involved in the medical event.",
  "treatment": "What therapy was administered to the patients?",
  "subject-race": "What is the race or nationality of the subject?",
  "subject-disorder": "What preexisting conditions does the subject have besides the main treatment disorder?"
}
```

Output JSON:

```
{ "questions": ["your set of questions"] }
```

C.3 Question Refinement

You are a helpful assistant who refines questions to better extract ground truth arguments for a given role. You previously generated a set of questions, but the extraction results show room for improvement. Your task is to analyze the feedback and produce an improved set of questions.

Context

Role: {role}

Document: {document}

Ground Truth Arguments: {gt_arguments}

Previous Questions (Iteration {iteration})

{current_questions}

Extraction Performance

Precision: {precision} **Recall:** {recall} **F1-Score:** {f1_score}

Summary:

- Successfully extracted: {n_matched_gt} out of {n_gt} ground truth arguments
- Total predictions: {n_pred} arguments
- Correct predictions: {n_matched_pred} out of {n_pred}

What Was Matched

These arguments were successfully extracted:
{matched_arguments}

What Was Missed

These ground truth arguments were NOT extracted:
{missing_arguments}

What Was Over-Generated

These arguments were predicted but are NOT in the ground truth (hallucinations or irrelevant extractions):
{extra_arguments}

Your Task

Based on the feedback above, **refine the question set** to:

1. **Add or modify questions** to capture the missing arguments
2. **Refine or remove questions** that led to over-generation or hallucinations
3. **Keep questions** that successfully extracted matched arguments (unless they also caused over-generation)
4. **Ensure clarity and specificity**—vague questions often lead to irrelevant extractions

Analysis Strategy

For Missing Arguments:

- Ask: “What specific question would directly prompt extraction of this argument from the document?”
- Consider: Is the information in the document? If yes, what keywords or phrases would target it?
- Action: Add a new targeted question OR modify an existing question to be more comprehensive

For Over-Generated Arguments:

- Ask: “Which question caused this hallucination or

irrelevant extraction?”

- Consider: Is the question too broad, ambiguous, or leading?
- Action: Make the question more specific, add constraints, or remove it if redundant

For Matched Arguments:

- Ask: “Which question successfully extracted this?”
- Action: Keep it, but check if it also caused over-generation

Guidelines:

- Avoid yes/no questions
- Keep questions concise and unambiguous
- Do not create redundant questions—each question should have a clear, distinct purpose
- The total number of questions does not have to stay the same; focus on **quality over quantity**

Critical Rules

Absolutely forbidden (violation makes questions useless for generalization):

1. Copying specific facts from “What Was Missed” into questions
2. Turning missed information into yes/no questions
3. Embedding document-specific details in questions
4. Asking “Did [specific fact] happen?”
5. Leaking ground-truth arguments through generated questions
6. Any other disguised leakage

Required:

1. Questions must target information *categories*, not specific values
2. Questions must be generic enough to work on different documents
3. Questions must be clear about what *type* of information is wanted

Output Format

Return your response **strictly in JSON format** as:
{“questions”: [“your refined set of questions”], “reasoning”: “Brief explanation of key changes and why”}

{“Age-at-Presentation”: “arg1; arg2”}.

Output JSON (use the role value from “Role to extract” as the key):
{ “{role}”: “arg1; arg2” }

D Auto-Interpretability

Below we give examples of automatically interpreting how questions transform using the matryoshka methodology from [Cherep et al. \(2026\)](#). We use an LLM-as-judge to label pairs of (Vanilla, Zero-Shot) or (Zero-Shot, Optimized) questions to interpret what changes, agglomeratively clustering embeddings of these differences and summarizing the clusters to the top-level summaries shown below. “Lexical” below means with token-level edit operations; otherwise diffs come from LLM-as-judge.

C.4 Prediction

You are a helpful assistant who **concisely** extracts arguments for a specified role from a given document, following the role question.

Role to extract: {role}

Role-specific Questions: {role_question}

Document: {document}

Instructions:

- Extract only the arguments relevant to the given role from the document following the role question.
- If no argument is present, return “null”.
- If multiple arguments are present, separate them with a semicolon (;).
- Do not include explanations, reasoning, or any extra text.
- Return the output strictly in JSON format.
- Do not repeat any part of the question.
- **The JSON must have exactly one key: the role name from “Role to extract” above. Do NOT use the word “role” as the key.** Example: if Role to extract is “Patient”, output {“Patient”: “arg1; arg2”}. If Role to extract is “Age-at-Presentation”, output

#	Document Excerpt	Vanilla Questions	List of Questions
1	<i>[CaseReportBench, Neuro]</i> On general examination, the patients were moderately built with a steady gait . The physical and mental development was also normal.	What neurological signs, symptoms, or findings are mentioned (e.g., motor skills, reflexes, sensation, mental status)?	Q1. How was the patient’s gait described in the general examination? Q2. What was reported about the patient’s mental development?
2	<i>[CaseReportBench, Lab/Imaging]</i> PSA was 0.393 ng/mL [...] X-ray of pelvis showed punctate calcification [...] CT scan showed extensive calcified areas [...] Biochemical analysis of urine showed calcium.	What laboratory test results or imaging findings are reported (e.g., MRI, CT, X-ray, biochemical tests)?	Q1. Was a prostate specific antigen (PSA) level reported? Q2. What calcification findings were described on the pelvic X-ray? Q3. What calcification findings were described on the CT scan? Q4. What substance was identified in the urine biochemical analysis? Q5. What components were identified in the biochemical analysis of the extracted prostatic calculi?
3	<i>[PHEE, Effect]</i> A 31-year-old man with OCD developed RBD soon after starting fluoxetine therapy, which persisted at PSG study 19 months after fluoxetine discontinuation.	What are the outcomes or side effects of the treatments?	Q1. What sleep disorder did the patient develop after starting fluoxetine therapy? Q2. According to the polysomnography study, how many months after fluoxetine discontinuation did this disorder persist?
4	<i>[PHEE, Treatment-Drug]</i> We report a case of recurrent torsades de pointes following treatment with pentavalent antimonial drugs and amiodarone .	Where the the drugs used as therapy in the event?	Q1. What antiarrhythmic drug was administered to the patient? Q2. What antimonial medication was administered to the patient?
5	<i>[DiscourseEE, Tapering-Type]</i> Post: I’m currently seeing an addiction doctor [...] they think withdrawals last only 3 days [...] I’m wondering if a family doctor can prescribe comfort meds?	Is the subject self-tapering or following prescribed tapering?	Q1. What instructions did the addiction doctor give about tapering the medication? Q2. What are the patient’s plans for handling the tapering of their medication?
6	<i>[DiscourseEE, Tapering-Event]</i> Post: I took 6 mgs then 4 mgs then 2 mg’s yesterday. It was closer to 2½ days [...] I don’t plan to take anymore [...] will I have any Suboxone withdrawals?	What are the tapering steps (drugs, start dosage, duration, goal dosage)?	Q1. What is the duration of the Suboxone taper before the individual stopped? Q2. What dosages of Suboxone were taken during the taper and in what order? Q3. What concerns does the individual express about potential withdrawal? Q4. What does the individual intend to do regarding Suboxone use after the taper?
7	<i>[MACCROBAT, Lab_value]</i> Admission blood test showed a slight decrease in hemoglobin (106 g/L).	What were the test results, measurements, or laboratory values from the diagnostic procedure?	Q1. What is the hemoglobin concentration reported in the admission blood test? Q2. How is the hemoglobin level described qualitatively?
8	<i>[MACCROBAT, Detailed_description]</i> The patient underwent 6 weeks of image-guided external beam radiation centred on the resection field.	What are the specific details or characteristics of the therapeutic procedure?	Q1. What type of radiation therapy was administered? Q2. What imaging technique was used to guide the radiation treatment?
9	<i>[MACCROBAT, Dosage]</i> She was given [...] sodium bicarbonate (8.4% , 10 flacon/day) infusion for 24 hours before chemotherapy.	What is the dosage, amount, or quantity of the medication?	Q1. What is the concentration of the sodium bicarbonate infusion administered? Q2. What is the daily amount of sodium bicarbonate given?

Table 12: Examples of LoQ questions compared to the Vanilla-Q counterparts across datasets and roles. Bold text in the document excerpt marks the distinct arguments each optimized question targets. For simple multi-argument roles (rows 7–9), LoQ generates 2 targeted questions; for complex roles it generates a set of questions to extract each argument correctly.

Dataset	Vanilla → Zero-Shot (LLM)	Zero-Shot → Optimized (Lexical)	Zero-Shot → Optimized (LLM)
CRB	• Scope narrowing to clinical features – Focus shifts from broad symptoms/signs to precise neurological, anatomical, ocular, or vital • Addition of temporal and treatment context – Inclusion of timing, treatment, or situational clinical scenarios in questions • Scope narrowing to specific entities and terminology – From broad categories to particular therapies, disorders, tests; includes terminology generalization • Question format and response constraint changes – Shift from open-ended to yes/no formats and addition of temporal constraints	• Lexical substitution and refinement – Replacing words or phrases with more precise, formal, or clinically focused alternatives • Insertion of clarifying or specifying terms – Adding words to specify context, timing, case references, or details • Deletion of redundant or less relevant text – Removing unnecessary words or phrases to streamline questions • Grammatical and structural adjustments – Singular-plural form changes and question phrasing or temporal expression modifications	• Clinical scope expansion – Adds symptoms, diagnostics, family history, outcomes, interventions with risk • Anatomical and biochemical specification – Includes brain regions, metabolites, lab findings, detailed anatomical features • Terminology refinement and formalization – Refines clinical/genetic terms, clarifies inheritance, replaces informal terms • Question format and wording adjustments – Shifts query style, splits questions, adds adjectives, clarifies timing and sources
DEE	• Subject perspective and terminology shifts – Changes between third-person and second-person subjects; 'subject' to 'patient' terminology • Temporal focus adjustments – Shifts from specific past/future events to broader or current time frames; additions • Scope narrowing and broadening – Changes between specific medication/substance and general drug categories or treatment status; shifts • Question format and focus shifts – Changing question types (yes/no to open-ended) and shifting emphasis among tapering, dosage	• Medication term generalization – Replacing specific drug names and dosages with general medication or treatment terms • Dosage and frequency phrasing changes – Modifying how dosage amounts and intake frequency are expressed or queried • Lexical substitution for question framing – Replacing interrogatives with descriptive or indirect wording and adding procedural details • Symptom and condition lexical refinement – Replacing general terms with specific symptom phrases and adjusting risk and temporal	• Scope broadening or narrowing – Changes in question focus between specific drugs/dosages and broader medication categories • Comparison target adjustment – Shifts in substances, conditions, or subjects compared or referenced • Temporal specification and perspective shifts – Introduction or refinement of timeframes and changes in temporal viewpoint • Supervision and tapering plan constraints – Includes lack of medical supervision and detailed tapering plan elements; medication type
MBAT	• Scope narrowing to specific clinical details – From general tests, procedures, and symptoms to precise measurements, assays, imaging features, • Disease and symptom specificity – General disease terms replaced with specific diseases; symptoms refined with situational, temporal, • Medication detail specification – Specifies exact medication names or classes, dosage, regimen, treatment context, and temporal • Question format and context shifts – Addition of temporal constraints, patient context, and shift to closed yes/no or	• Lexical substitution for specificity – Replacing vague terms with precise clinical or descriptive words • Interrogative form modification – Changing question words, phrasing, structure, and punctuation including synonyms • Lexical insertion and deletion – Adding or removing descriptive, temporal, or passive terms • Question phrasing shift – Changing open-ended questions to closed or explicit formats, including modality changes	• Numeric to qualitative shift – From exact numeric values to qualitative or evaluative answers • Open-ended to closed-ended format – Changed question style from open response to yes/no format • Terminology refinement and generalization – Rephrasing specific terms to broader or more precise descriptors, including technical terms • Scope modification and narrowing – Changes in entity focus, timing, anatomy, symptom triggers; removal of patient references;

Dataset	Vanilla → Zero-Shot (LLM)	Zero-Shot → Optimized (Lexical)	Zero-Shot → Optimized (LLM)
PHEE	• Patient population specificity – General individual references changed to specific patient groups or conditions • Specified medical entities – Medication or condition explicitly named (e.g., ampicillin, retinopathy, meningitis) • Shift to interrogative questions – Changed from general statements or noun phrases to explicit question forms • Disorder category consolidation and wording simplification – Combining/reducing disorder categories; removing explanatory clauses; replacing 'subject' with 'patient(s)'; simplifying phrasing • Lexical substitution and insertion – Replacing or adding terms to refine focus or increase specificity • Insertion of qualifiers and descriptors – Adding adjectives, numeric qualifiers, or clinical details to entities • Addition of clarifying phrases – Including explicit question components or clarifying clauses • Specification of subject and condition – Clarifying species, cases, or condition references within questions • Patient group focus shifts – Changes in referenced patient populations and clinical attribute emphasis • Adverse effect and therapy specificity – Identification of specific adverse effects and explicit drug or agent naming • Causation versus association framing – Shift from general therapy association to specific causative agent inquiry • Scope modification and question clarity – Scope narrowing or broadening by disorder/treatment specificity and qualifier removal; question wording		